

DSP Lab - MATLAB Codes

Experiments 01 to 10 (code only)

Experiment No. 01

Aim: Write a MATLAB code and plot DFT of a sequence without using the inbuilt functions.

```
clc;
clear all;
close all;
xn = input('enter the sequence x(n)');
ln = length(xn);
xk = zeros(1,ln);
for k=0:ln-1
    for n=0:ln-1
        xk(k+1) = xk(k+1) + (xn(n+1))*exp((-i)*2*pi*k*n/(ln));
    end
end
t = 0:ln-1;
subplot(221);
stem(t,xn)
grid;
ylabel('Amplitude')
xlabel('time')
title('input sequence');
magnitude = abs(xk);
subplot(222);
stem(t,magnitude);
ylabel('Amplitude');
xlabel('k');
title('magnitude response');
phase = angle(xk);
subplot(223);
stem(t,phase);
ylabel('phase');
xlabel('k');
title('phase response');
```

Experiment No. 02

Aim: Write a MATLAB code and plot IDFT of a sequence without using the inbuilt MATLAB functions.

```
clc
xn = input('Enter the sequence xn')
ln = length(xn)
xk = zeros(1,ln)
ixk = zeros(1,ln)
for k=0:ln-1
    for n=0:ln-1
        xk(k+1) = xk(k+1) + (xn(n+1)*exp(-i*2*pi*k*n/(ln)))
    end
end
disp(xk)
t = 0:ln-1
subplot(2,2,1)
stem(t,xn)
grid
xlabel('time domain index')
ylabel('Amplitude')
title('input sequence')
magnitude = abs(xk)
t = 0:ln-1
subplot(2,2,2)
stem(t,magnitude)
grid
ylabel('Amplitude')
xlabel('k')
title('magnitude Response')
phase = angle(xk)
t = 0:ln-1
subplot(2,2,3)
stem(t,phase)
grid
xlabel('k')
ylabel('phase')
title('phase Response')
for n=0:ln-1
    for k=0:ln-1
        ixk(n+1) = ixk(n+1) + (xk(k+1)*exp(i*2*pi*k*n/(ln)))
    end
end
ixk = ixk/ln
t = 0:ln-1
subplot(2,2,4)
stem(t,ixk)
grid
ylabel('Amplitude')
xlabel('Time Index')
title('IDFT sequence')

% Output:
% Enter the sequence xn: [10 -2+2j -2 -2-2j]
```

Experiment No. 03

Aim: Write a MATLAB code & plot circular convolution of given sequence using MATLAB function.

```
clc
a = input('Enter the sequence x(n) =')
b = input('Enter the sequence h(n) =')
n1 = length(a);
n2 = length(b);
N = max(n1,n2)
x = [a zeros(1,(N-n1))]
for i=1:N
    k = i
    for j=1:n2
        H(i,j) = x(k)*b(j)
        k = k-1
        if (k==0)
            k = N
        end
    end
end
y = zeros(1,N)
m = H
for j=1:N
    for i=1:n2
        y(j) = m(j,i) + y(j)
    end
end
disp('The output sequence is y(n)=');
subplot(1,3,3)
disp(y)
stem(y)
xlabel('n')
ylabel('y(n)')
title('circular convolution')
subplot(1,3,1)
stem(a)
title('input x(n)')
xlabel('n')
ylabel('x(n)')
subplot(1,3,2)
stem(b)
title('input h(n)')
xlabel('n')
ylabel('h(n)')

% COMMAND WINDOW INPUT:
% Enter the sequence x(n) = [1 2 3 1]
% Enter the sequence h(n) = [4 3 2 2]
% OUTPUT:
% The output sequence is y(n): 17 19 22 19
```

Experiment No. 04

Aim: Write a MATLAB code to apply FFT on a given sequence and plot the magnitude and phase response of the same.

```
clc
n = input('enter the length of input:')
t = 0:1:n
y = input('Enter the input sequence:')
subplot(3,1,1)
stem(y)
grid
title('input sequence')
xlabel('time')
ylabel('amplitude')
disp(y)
xn = y
N = input('Enter the length of the FFT')
xk = fft(xn,N)
magxk = abs(xk)
angxk = angle(xk)
k = 0:N-1
subplot(3,1,2)
stem(k,magxk)
grid
xlabel('k')
ylabel('|x(k)|')
title('magnitude response')
subplot(3,1,3)
stem(k,angxk)
grid
disp(xk)
xlabel('k')
ylabel('arg(x(k))')
title('angular response')

% COMMAND WINDOW INPUT:
% Enter the input sequence: [1 2 3 4]
% OUTPUT:
% output sequence: [10 -2+2j -2 -2-2j]
```

Experiment No. 05

Aim: Write a MATLAB code to apply IFFT on a given sequence and plot the magnitude and phase response of the same.

```
clc
clear all
close all
x = input('Enter the sequence of freq. domain');
ln = length(x)
t = 0:ln-1
y = ifft(x)
mag = abs(y)
phase = angle(y)
disp('Output sequence')
disp(y)
subplot(2,2,1)
stem(x)
xlabel('k')
ylabel('Amplitude')
title('Input sequence')
subplot(2,2,2)
stem(t,mag)
xlabel('time')
ylabel('Amplitude')
title('magnitude Response')
subplot(2,2,3)
stem(t,phase)
xlabel('time')
ylabel('Amplitude')
title('phase Response')

% Output:
% Enter the sequence of freq. domain [10 -2+2j -2 -2-2j]
% Output sequence in time domain is [1 2 3 4]
```

Experiment No. 06

Aim: Write a MATLAB code to implement low pass FIR filter for given sequence.

```
clc
wc = 0.5*pi
N = 5
alpha = (N-1)/2
eps = 0.001
n = 0:1:N-1
hd = sin(wc*(n-alpha+eps))./(pi*(n-alpha+eps))
wr = boxcar(N)'
hn = hd.*wr
w = 0:0.1:pi
h = freqz(hn,1,w)
plot(w/pi,abs(h))
hold on
plot(w/pi,abs(h))
grid
ylabel('magnitude')
xlabel('normalised frequency')
hold off
```

Experiment No. 07

Aim: Write a MATLAB code to implement high pass FIR filter for given sequence.

```
clc
wc = 0.5*pi
N = 5
alpha = (N-1)/2
eps = 0.001
n = 0:1:N-1
hd = (sin(pi*(n-alpha+eps)) - sin(wc*(n-alpha+eps)))./(pi*(n-alpha+eps));
wr = boxcar(N)'
hn = hd.*wr
w = 0:0.1:pi
h = freqz(hn,1,w)
plot(w/pi,abs(h))
hold on
plot(w/pi,abs(h))
grid
ylabel('magnitude')
xlabel('normalised frequency')
hold off
```

Experiment No. 08

Aim: Write a MATLAB code to implement Band pass FIR filter for given sequence.

```
clc
clear all
wc1 = 0.25*pi
wc2 = 0.75*pi
N = 5
alpha = (N-1)/2
eps = 0.001
n = 0:1:N-1
hd = (sin(wc2*(n-alpha+eps)) - sin(wc1*(n-alpha+eps)))./(pi*(n-alpha+eps))
wr = boxcar(N)'
hn = hd.*wr
w = 0:0.01:pi
h = freqz(hn,1,w)
plot(w/pi,abs(h))
ylabel('magnitude')
xlabel('normalised frequency')
```

Experiment No. 09

Aim: To design a Hilbert transform function using Bartlett and Hamming windows.

```
clc
N = 5
alpha = (N-1)/2
eps = .001
n = 0:1:N-1
hd = (1-cos(pi*(n-alpha+eps)))./(pi*(n-alpha+eps))
hd(alpha+1) = 0
wt = bartlett(N)
hn = hd.*wt
w = 0:.01:pi
h = freqz(hn,1,w)
plot(w/pi,abs(h))
hold on
wh = hamming(N)
hn = hd.*wh
w = 0:0.1:pi
h = freqz(hn,1,w)
plot(w/pi,abs(h),'-.')
ylabel('magnitude')
xlabel('normalised frequency')
hold off
```

Experiment No. 10

Aim: To design a Butterworth low pass filter for given specification.

```
clc
alphap = 0.4
alphas = 30
fp = 400
fs = 800
F = 2000
omp = 2*fp/F
oms = 2*fs/F
[n,wn] = buttord(omp,oms,alphap,alphas)
[b,a] = butter(n,wn)
w = 0:0.01:pi
[h,om] = freqz(b,a,w)
m = 20*log10(abs(h))
an = angle(h)
subplot(2,1,1)
plot(om/pi,m)
grid
ylabel('gain in db')
xlabel('normalised frequency')
subplot(2,1,2)
plot(om/pi,an)
grid
ylabel('gain in radians')
xlabel('normalised frequency')
```